

Blood neurofilament (NfL) vs ALSFRS-R decline in PRO-ACT

H-002 - baseline serum NfL tracks faster ALSFRS-R decline ($\rho = -0.43$, $n = 788$); survival not testable

Owner: Manu Dataset: PRO-ACT (data/PROACT_ALL_FORMS) Date: 2026-09-07

Headline: higher baseline serum NfL -> faster ALSFRS-R decline (Spearman $\rho = -0.43$, 95% CI [-0.48, -0.37], $n = 788$)

Status: H1 supported (survived 4 adversarial lenses); H2 survival not testable

Reproducibility: .venv/Scripts/python.exe scripts/analysis.py (claims.json gate: 14/14)

PRIMARY CLAIM SUPPORTED (SURVIVED 4 INDEPENDENT REVIEW LENSES); SURVIVAL NOT TESTABLE; TWO SECONDARY OVER-READS DEMOTED

In PRO-ACT, higher baseline serum NfL goes with faster subsequent ALSFRS-R decline (Spearman $\rho = -0.43$, 95% CI [-0.48, -0.37], $n = 788$, permutation $p = 2e-4$). It holds after covariate adjustment and precision-weighting, and the survivorship lens shows the selection biases it toward zero, so -0.43 is a conservative floor. It replicates the direction of the motivating plasma preprint ($r = -0.53$), attenuated as expected for serum. Three secondary claims were demoted by adversarial review: the regression-slope magnitude (specification-dependent), the 61 pg/mL cutoff (a re-expression, not independent evidence), and the survival signal (a collider-biased, decedent-only sample - not testable here). This is observational cohort data in a limb-enriched, faster-progressing subgroup, not ALS at large.

-0.43

Spearman: baseline serum NfL vs ALSFRS-R slope

95% CI [-0.48, -0.37], $n = 788$, permutation $p = 2e-4$. Survived confounding, survivorship, statistics, reproducibility.

788

Subjects with serum-baseline NfL AND a fittable slope

of 814 with any serum NfL. A limb-onset-enriched, faster-progressing subgroup - the trials that assayed NfL.

0 / 114

Censored subjects in the NfL-survival overlap

All 114 linked NfL subjects had died. Survival is decedent-only, so H2 is not testable in PRO-ACT.

Hypotheses and status

#	Claim	Status
H1	Higher baseline serum NfL -> faster ALSFRS-R decline slope	Supported ($\rho = -0.43$, $n = 788$)
H1b	The 61 pg/mL split separates fast from slow decliners	Demoted - re-expression of H1, not independent evidence
H2	Higher baseline NfL -> shorter survival	Inconclusive - not adequately testable (0 censored)

Data and hypothesis

Data - PRO-ACT, de-identified, workspace-resident

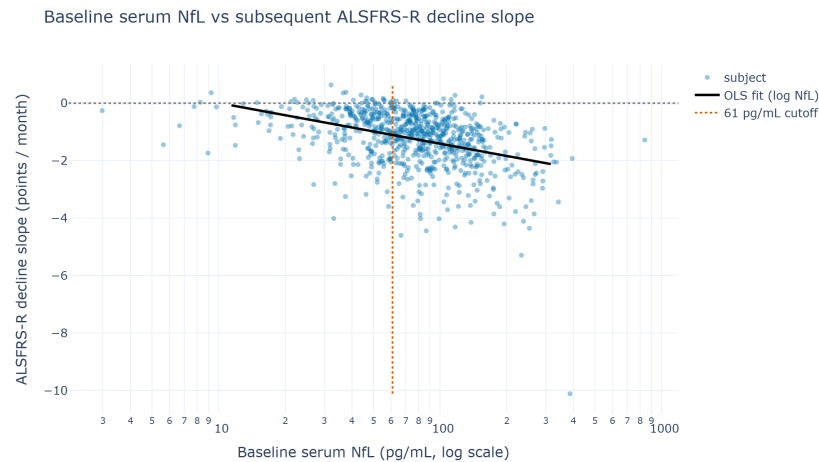
- **F_PROACT_Neurofilament** - serum NfL (pg/mL); baseline = earliest draw within 0-90 days. Median 78.5 pg/mL (IQR 52.6-115.1, max 840). Serum only (CSF excluded).
- **F_PROACT_ALSFRS** - ALSFRS_R_Total over time -> per-subject OLS slope (≥ 3 visits, ≥ 90 -day span, first 18 months; identical to H-001).
- **F_PROACT_DEATHDATA / DEMOGRAPHICS / ALSHISTORY** - survival and covariates (age, sex, onset, disease duration, baseline ALSFRS-R).
- Join on subject_id. **Primary cohort n = 788**. Riluzole dropped (no "No" rows in this cohort -> a trial-presence flag, not a treatment contrast).

Hypothesis and motivation

H1: baseline serum NfL correlates with the ALSFRS-R decline slope (negative). **H2 (exploratory):** NfL predicts survival.

Motivation: a medRxiv PREPRINT (not peer reviewed), Arguedas et al. (ALS/MND NHC, $n = 300$), reporting plasma NfL vs ALSFRS-R rate of change $r = -0.53$ with a 61 pg/mL cutoff. Tested here as a hypothesis on an independent, larger, serum cohort - not treated as an established finding.

Finding F1 - Higher baseline serum NfL tracks faster ALSFRS-R decline (SURVIVES)



Baseline serum NfL (log scale) vs subsequent ALSFRS-R decline slope; the black line is the OLS fit on log-NfL, the dotted line marks 61 pg/mL.

SURVIVED ALL FOUR ADVERSARIAL LENSES

Spearman rho = -0.43 (95% CI [-0.48, -0.37], n = 788). Holds adjusted for age, sex, onset, disease duration, baseline ALSFRS-R (beta = -0.47, n = 365; -0.51 on the less-selected n = 734) and under precision-weighting (beta = -0.46). Partial r = -0.35 controlling disease duration - NfL is not merely a proxy for how early the patient enrolled.

CLINICAL / MEDICAL

NfL is an axonal-cytoskeleton protein released into blood as motor-neuron axons degenerate, so it indexes the rate of axonal destruction. Higher baseline NfL marking steeper ALSFRS-R decline is coherent - but NfL and functional loss are two readouts of the same process, so this is a shared-cause association, not proof NfL drives progression.

DATA-SCIENCE / STATISTICS

NfL is right-skewed (median 78.5, max 840 pg/mL), so the rank-based Spearman is the headline and log-NfL is used parametrically. rho = -0.43 is a moderate effect (~18% of rank variance); it is foregrounded over the vanishing p because large n makes trivial effects extreme. CI is a 2000-sample bootstrap of pairs.

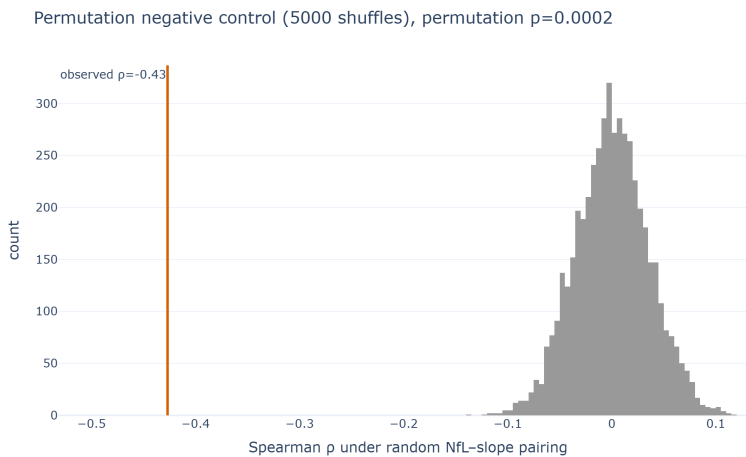
PLAIN ENGLISH

NfL is like debris shed from fraying electrical wiring. The more debris in the blood at the start, the faster the wiring came apart - patients lost function (speech, walking, breathing, hand use) faster over the next 18 months. A tendency across a crowd, not a verdict for one person.

CONCRETE DATA EXAMPLE

Subject 7527493 (baseline serum NfL 211 pg/mL) declined at -2.15 ALSFRS-R points/month; subject 2963489 (NfL 25.5 pg/mL) was essentially stable (about 0 points/month). Two ends of the same gradient.

Finding F2 - The association is not an artifact of chance pairing (SURVIVES)



Permutation negative control: the null distribution of Spearman rho from 5000 shuffles sits at 0; the observed rho = -0.43 (orange line) is far outside it.

PERMUTATION NEGATIVE CONTROL - OBSERVED EFFECT FAR OUTSIDE THE NULL

Shuffling the NfL-slope pairing 5000 times gives a null centred on 0 (mean 0.0005, SD 0.035). The observed rho = -0.43 lies entirely outside the null (not one of 5000 shuffles reached it), permutation $p = 2e-4$.

CLINICAL / MEDICAL

This adds no biology; it protects the claim, confirming the link reflects each patient's own NfL matched to their own decline rather than a coincidence of two distributions.

DATA-SCIENCE / STATISTICS

Permutation breaks the NfL-slope dependency while preserving both marginals, giving the exact null for "no association." The observed statistic sits about 12 null-SDs out - the required negative control on the headline effect (team rule 3).

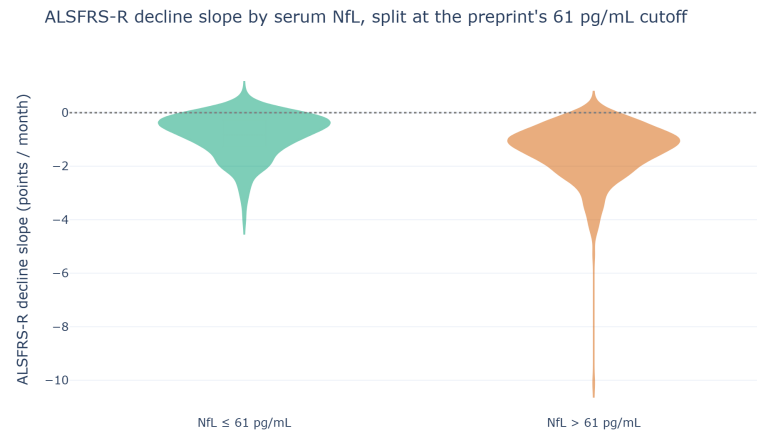
PLAIN ENGLISH

We randomly re-matched each patient's marker to a different patient's decline 5000 times. Random matching essentially never produced a link this strong, so the real link is very unlikely to be a fluke of the numbers.

CONCRETE DATA EXAMPLE

A representative shuffle of the same 788 values yields rho about +0.01 (indistinguishable from zero); the true pairing yields rho = -0.43.

Finding F3 - The 61 pg/mL split (DEMOTED: a re-expression, not independent evidence)



ALSFRS-R decline slope split at the preprint's 61 pg/mL threshold: subjects above the line decline faster than those at or below it.

KEPT ONLY AS AN ILLUSTRATION OF FINDING 1 - NOT A SECOND, INDEPENDENT RESULT

Above 61 pg/mL (n = 520) mean slope -1.44 points/month vs -0.84 at or below (n = 268). This is Finding 1 re-expressed as a binary split, not a second confirmation; the log-NfL x group interaction is a collinearity artifact (the high/low flag is a deterministic step of log-NfL) and must not be read as effect modification.

CLINICAL / MEDICAL

A dichotomous threshold is clinically attractive for prognosis and trial enrichment, and a serum threshold of this order does separate fast from slow progressors here. But the 61 pg/mL value came from plasma; serum on a different assay is not interchangeable, so read it as "high vs low," not as validation of 61 pg/mL.

DATA-SCIENCE / STATISTICS

Dichotomizing a continuous predictor discards information and inflates group contrast; it is kept only as an interpretable picture of Finding 1. The cutoff is imported, not fitted here, so it is not circular - but it is not extra evidence either.

PLAIN ENGLISH

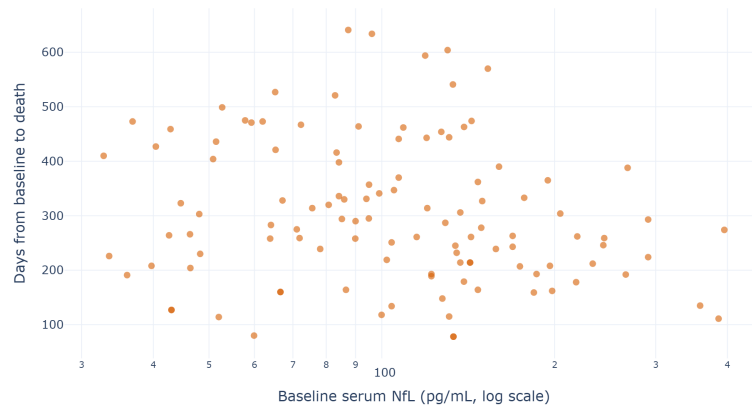
If we draw a line at the marker level the preprint used, people above it lost function faster. It is the same finding shown as two groups, and the exact line depends on the lab test.

CONCRETE DATA EXAMPLE

Subject 7527493 (211 pg/mL, above the line) declined -2.15 points/month; subject 2963489 (25.5 pg/mL, below) was stable - the same two subjects, viewed as high vs low.

Finding F4 - NfL and survival (DEMOTED: not testable in PRO-ACT)

EXPLORATORY (n=114, all deaths, no censoring): baseline NfL vs time to death



Exploratory only (n = 114, all deaths, no censoring): baseline serum NfL vs days from baseline to death.

NOT A FINDING - DECEDENT-ONLY LINKAGE SELECTS ON THE OUTCOME

Only 114 subjects have both a baseline serum NfL and a death record, and all 114 died (0 censored); 697 NfL subjects have no death record at all. Any model here selects on the outcome. A Cox restatement gives HR = 1.24 per SD of log-NfL (95% CI [1.00, 1.54]) - null on its own terms (p = 0.053) and not a mortality hazard ratio.

CLINICAL / MEDICAL

Higher NfL plausibly predicts shorter survival (faster axonal loss -> earlier respiratory failure) and the sign is consistent, but with survivors structurally absent this cannot be estimated without bias.

DATA-SCIENCE / STATISTICS

Cox regression needs censored observations to estimate the hazard for survivors; with 0/114 censored the risk set never contains a survivor, so the estimate is conditioned on having died. The within-decedent rank association (Spearman -0.19, p = 0.04) is all that is computable.

PLAIN ENGLISH

We wanted to ask whether the marker predicts how long people live, but for almost everyone with the marker we only have records of those who already died - no fair comparison group of survivors. The hint points the expected way, but we cannot trust the number or even its direction.

CONCRETE DATA EXAMPLE

Subject 4506836 (baseline NfL 218 pg/mL) died 178 days after baseline - one point consistent with the trend, but 114 such points, all deaths, cannot anchor a survival model.

What survived the adversarial review

H1 - the primary rank association held on all four lenses

- **Confounding:** survives adjustment and a partial correlation controlling disease duration ($r = -0.35$); NfL is not a mere severity proxy.
- **Survivorship:** the selection filters remove high-NfL fast progressors, biasing rho *toward zero* - so -0.43 is a conservative underestimate.
- **Statistics:** correct estimators (paired bootstrap, Fisher-z), no circularity, one subject one row.
- **Reproducibility:** byte-identical re-runs; all 14 headline numbers tie to the pipeline via claims.json.

What was withdrawn or demoted (so no one re-derives it)

Demoted / withdrawn claims, with the mechanism that defeated each

- **"Adjusted for riluzole"** - WITHDRAWN. The riluzole form has no "No" rows here, so it encoded trial-presence, not treatment. Removed from the model.
- **61 pg/mL cutoff as independent corroboration** - DEMOTED to a re-expression of H1; the interaction term is a collinearity artifact.
- **"Cox HR = 1.24 predicts survival"** - DEMOTED to a collider-biased within-decedent rank association; null on its own terms.
- **Data-section "max 27,212 pg/mL"** - CORRECTED to 840; the large value was an excluded CSF outlier from the recon step, not the analyzed serum variable.
- **beta = -0.62 per log-unit as a firm magnitude** - QUALIFIED; precision-weighting moves it to -0.46, so the magnitude is specification-dependent (the correlation is not).

Comparison to the motivating preprint

Aspect	Preprint (medRxiv, NOT peer reviewed)	This analysis (PRO-ACT)
Matrix	Plasma	Serum
n (correlation)	300	788
NfL vs ALSFRS-R rate	$r = -0.53 [-0.62, -0.42]$	$\rho = -0.43 [-0.48, -0.37]$
61 pg/mL cutoff	association differs at cutoff	present on serum, assay-dependent, not independent evidence
Survival	reported predictive	not adequately testable (no censoring)

DIRECTION REPLICATES; MAGNITUDE ATTENUATES

Consistent with serum-vs-plasma matrix, assay differences, and a limb-enriched cohort. A genuinely new test of the claim on an independent, larger cohort - not a duplication.

Limitations

Read the headline within these bounds

- **Observational** - correlation, not causation; NfL and ALSFRS-R decline share a cause.
- **Non-representative cohort** - limb-enriched, near-bulbar-free, faster-progressing; a single/few-substudy frame, so absolute NfL and the 61 pg/mL cutoff are likely assay/trial-specific.
- **Serum not plasma**; NfL exists for only ~8% of PRO-ACT.
- **beta magnitude is specification-dependent** (generated-regressor structure); the rank correlation is robust, the coefficient less so.
- **Survivorship in the slope** under-represents the fastest progressors, biasing H1 toward zero (conservative).
- **Survival not testable** - decedent-only linkage, no censoring.

Open questions

Q1

Does the association hold in bulbar-onset ALS?

This cohort is almost entirely limb-onset. Bulbar disease progresses differently, so the result may not transfer.

Q2

Can a longitudinal NfL slope beat baseline NfL?

798 subjects have ≥ 2 serum draws. A within-subject NfL trajectory might predict decline better than a single baseline value.

Q3

Is the serum-to-plasma cutoff mapping recoverable?

The 61 pg/mL threshold is plasma-derived; a serum-equivalent would need a cross-platform calibration this dataset does not carry.

Reproducibility - runs end-to-end on a workstation

- **analysis.py** (end-to-end) + **recon.py** (data profiling) in `scripts/`; **claims.json** standing gate (14/14 numbers tied to the pipeline).
- **4 figures** (HTML + PNG), **results.json**, `subject_level.csv` (local only).
- Run: `.venv/Scripts/python.exe scripts/analysis.py`. Determinism: RNG seed 20260907 for the bootstrap CI and the 5000-shuffle permutation; re-runs are byte-identical.